PRODUCTION DOCUMENT | UI / UX SPECIFICATION

# AI Video Studio UI/UX Spec

What every screen shows, how each interaction behaves, and what happens when things are empty, loading, or broken. Written so a designer can draw it and a developer can build it without guessing.

> **The product in one line:** A web app (Windows + Mac, browser-based) that takes a creator from topic to a finished video plus a publish pack - Shorts / Reels, thumbnails, title and description - in one project, in one flow.

## How to use this spec

- **Decisions are final unless listed in Section 11.** Section 11 is the only place open choices live, each as one explicit decision with options and a recommended default.
- **Stage numbers are stable.** Screens, URLs, analytics events, and support tickets all refer to stages 1-10 by the names in Section 2.
- **Every AI action follows one pattern:** propose, preview, apply or reject, undo. No screen invents its own variant (Section 6.1).
- **Stack assumed throughout:** Next.js frontend, FastAPI orchestrator, Redis / Celery job queue, faster-whisper transcription, Remotion motion graphics and render, ffmpeg media processing, R2 / S3 asset storage.

## Scope for MVP

In: dashboard, project, script, voice and transcript, visual plan, timeline, captions and sound, preview, render at 1080p, publish pack. Out for MVP: 4K render, multi-user collaboration, public template marketplace, in-app publishing directly to YouTube or Instagram (download-and-upload only).

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Product: AI Video Studio | Companion docs: PRD, Architecture and Data, API Spec, AI Pipeline Spec, Security and Licensing, Testing and DevOps, Pricing and Launch
Asset rule: licensed stock (Pexels / Pixabay) is the default visual source. YouTube clip use is opt-in only, per project, with the risk warning in Section 6.2.
Pricing model reflected in UI: credits / subscription. Render and generation costs are shown before the user commits, never after.

# 1. Principles and information architecture

## 1.1 Five principles

1. **AI proposes, the user disposes.** Nothing AI-generated replaces user work silently. Every generation lands as a reviewable proposal first.
2. **The script is the source of truth.** Voice, visuals, captions, and graphics all trace back to a script block. Change the script and the UI shows exactly what downstream is now stale.
3. **Costs before commitment.** Credit cost is shown on the button that spends it ("Generate - 12 credits"), never only in a billing page.
4. **No dead ends.** Every loading, empty, and error state offers one clear next action.
5. **Reversible by default.** Undo covers at least the last 20 actions per project. Destructive actions (delete project, delete take) ask once and offer Undo for 10 seconds after.

## 1.2 Information architecture

| Level | Contents |
|---|---|
| Workspace (account) | Dashboard, Templates, Assets library, Billing and credits, Settings |
| Project | Overview, Script, Voice, Visuals, Timeline, Captions and Sound, Preview, Render, Publish |
| Global | Help, keyboard shortcuts overlay (?), job centre (active renders / generations), notifications |

One project = one long-form video plus its derived publish pack. Shorts are not separate projects; they live inside the parent project under Publish, so assets and credits stay in one place.

## 1.3 Roles in MVP

Single user per account. Owner sees everything including billing. No editor / viewer split in MVP; the data model must not block adding it later (projects carry an owner ID, not embedded user data).

# 2. Global shell and the 10-stage pipeline

## 2.1 Stage map

| # | Stage | Screen | Done when |
|---|---|---|---|
| 1 | Topic | Project > Overview | Topic, audience, target length saved |
| 2 | Research | Project > Overview | At least 1 source accepted into brief |
| 3 | Script | Script | Script approved (explicit click) |
| 4 | Voice | Voice | A take (recorded or AI) selected |
| 5 | Voice cleanup | Voice | Cleanup applied or explicitly skipped |
| 6 | Visual plan | Visuals | Every script block has a visual |
| 7 | Generation | Visuals | Assets generated into the project |
| 8 | Motion graphics | Timeline | Graphics placed or stage skipped |
| 9 | Timeline edit | Timeline | User clicks "Ready to preview" |
| 10 | Captions, sound, preview, render, publish | Captions / Preview / Render / Publish | Render complete; pack downloaded |

Stages are a guide rail, not a lock. The user can move backwards freely. Moving forward past an incomplete stage is allowed except Render, which requires: approved script, selected voice take, a visual on every block, and captions generated.

## 2.2 App shell (all project screens)

| Top bar: logo \| project name (editable) \| Saved 12:04 \| Credits: 340 \| Job centre (2) \| Avatar |
| --- |
| Left rail (stage list, 200px) \| Main canvas (flexible) \| Optional right inspector (320px, collapsible)<br><br>Bottom (Timeline screen only): timeline strip, 180-260px, resizable |

- **Stage rail:** each stage shows one of four states - not started (grey), in progress (dot), complete (tick), stale (amber dot: upstream changed). Stale is explained on hover and on the affected screen, never by colour alone.
- **Autosave:** every edit saves within 1 second (debounced). Top bar shows Saving / Saved HH:MM / Save failed - retry. No Save button anywhere.
- **Job centre:** all Celery jobs (transcription, generation, render) listed with progress, stage link, and Cancel where the backend supports it. Closing the tab never cancels a job.
- **Credits pill:** always visible inside a project. Click opens a breakdown: balance, cost of last job, estimated cost of next step.
- **Keyboard:** Ctrl/Cmd+Z undo, Ctrl/Cmd+Shift+Z redo, Ctrl/Cmd+S forces save checkpoint, ? opens shortcuts, Esc closes any panel or modal.

## 2.3 Project status values

Draft | In production | Rendering | Ready to publish | Published | Archived

Status is derived from stage completion, not set manually, except Archived (user action) and Published (user confirms after downloading the pack).

# 3. Dashboard

The dashboard has one primary job: get a returning user back into their last project in one click.

Dashboard

[ + New project ]  Search projects...  Sort: Recent | Name

Continue: [Last project card - thumbnail, stage 7 of 10, Edited 2h ago, Resume]
Grid: project cards (16:9 thumbnail, title, status, duration target, updated, credit spent)
Below fold: Templates (6 starters) | Recent renders

### DECISIONS

- **New project** opens a 3-field modal only: topic (required), target length (3 / 6 / 10 min presets or custom), format (16:9 default; 9:16-only project is an open decision, Section 11). Everything else is set inside the project. No 5-step wizard.
- **Card actions** (on hover / kebab menu): Open, Duplicate, Rename, Archive, Delete. Duplicate copies script and settings, not rendered files.
- **Empty state (new user):** headline "Make your first video", one primary button, and three starter topics the user can click to pre-fill - not sample projects presented as their own work.
- **Search** matches title and topic text. Filters: status only in MVP.
- Thumbnails: auto frame from the project once visuals exist; before that, a neutral block with the project initial - never a random stock image.

# 4. Project overview, research, and Script

## 4.1 Project overview (Stages 1-2)

- **Brief panel:** topic, hook (one line), audience, tone (pick from 5: Explainer, Documentary, Energetic, Calm, Direct), target length, language. All inline-editable.
- **Research panel:** AI research appears as source cards - title, source name, date, 2-line summary, and the claim it supports. User actions per card: Use in brief, Dismiss. Only "Use in brief" cards feed the script generator. Each card links to its source; no source, no card.
- **Regenerate research** adds new cards below existing ones; it never deletes accepted cards.

## 4.2 Script screen (Stage 3)

| Script \| Word count 812 \| Est. duration 6:10 \| [Approve script] |
| --- |
| Left: block list (Hook / Section / Example / CTA) - drag to reorder<br>Centre: block editor - title, narration text, on-screen text, duration estimate<br>Right inspector: tone check, reading time, research links used in this block |

- Script is a list of **blocks**, not one text blob. Each block maps 1:1 to voice segment, visual, and caption range downstream.
- **AI rewrite is per block:** Shorter, Punchier, Simpler, Rewrite with instruction. Result appears as a proposal diff (Section 6.1); original is never overwritten in place.
- **Approve script** is an explicit button. Editing an approved script un-approves it and marks Voice and Visuals stale if narration text changed.
- Word count and estimated duration update live (130 words/min default, adjustable per project in Voice settings).
- Empty state: "Generate a draft from brief" (shows credit cost) or "Write it myself" (creates 3 empty blocks: Hook, Main, CTA).

# 5. Voice, cleanup, and transcript editing

## 5.1 Two voice paths, one screen

| Record my voice | AI voice |
|---|---|
| Record per block or as one full take. Browser recording with level meter, countdown, pause / resume. Upload WAV / MP3 / M4A also accepted (max 500 MB). | Pick voice from a curated list with 10-second samples, speed (0.9-1.1x) and stability controls. Generate per block or full script. Cost shown per generation. |

## 5.2 Cleanup (Stage 5)

- Toggles, each previewable A/B on a 15-second sample before applying: remove silences over 0.6s, remove filler words (um, uh - transcript-detected), breath reduction, loudness normalise to -14 LUFS.
- Cleanup produces a **new take version**; the raw take is always kept and selectable. Applying cleanup twice never stacks - it always runs from raw.
- Progress via Job centre; a block-level spinner in the take list, not a full-screen blocker.

## 5.3 Transcript editing - the core interaction

- faster-whisper transcript appears word-timed under each block. Click a word to play from that word. Select words and press Delete to cut that audio (ripple delete). Edit a word's text to fix captions without touching audio.
- Two edit modes, clearly labelled: **Edit captions only** (default, safe) and **Edit audio by text** (cuts audio, requires confirm, fully undoable).
- Low-confidence words (under 0.6) are underlined amber with a one-click list: "Review 6 uncertain words".
- Speaker labels: single-speaker in MVP; the transcript UI must not show a diarisation control it cannot honour.

# 6. Visual plan, generation, and Timeline

## 6.1 The proposal pattern (applies everywhere)

1. AI action runs and returns a **proposal card**: what will change, a preview or diff, credit cost if applied (or "already paid" if generation cost was upfront).
2. User chooses **Apply**, **Try again** (new proposal, old one kept in proposal history), or **Dismiss**.
3. After Apply, a toast offers **Undo** for 10 seconds; full history remains in Edit > History (last 20 actions, each restorable).
4. Chat-style edit box (Timeline screen): the user types an instruction ("tighten the intro", "swap block 3 visual"); the system replies with a proposal card, never a silent edit. This borrows the propose-then-apply pattern seen in recent AI editors, implemented clean-room.

## 6.2 Visual plan (Stages 6-7)

- One row per script block: narration excerpt, visual type selector (Stock video / Stock photo / AI image / AI video / Screen-style graphic / Text card), prompt or search terms (editable), and the chosen asset thumbnail.
- **Default source: licensed stock** (Pexels / Pixabay). Licence label is shown on every asset (e.g. "Pexels - free to use") and stored with the project for the publish pack credits file.
- **YouTube clips: opt-in only.** Off by default, enabled per project in Settings with a plain-language warning that downloading clips can breach platform terms and copyright, and that the user is responsible for rights. The toggle, its warning text, and the timestamp of consent are specified in the Security and Licensing doc.
- Generate / search results appear as a 4-up pick list per block. Picking replaces; the previous pick stays in the block's asset history.
- Stage is complete only when every block has an asset or is explicitly marked "Text / graphic only".

### 6.3 Timeline editor (Stages 8-9)

| Preview monitor \| Playhead time \| Safe-area toggle \| Zoom |
| --- |
| Tracks: V2 Graphics \| V1 Video \| A1 Voice \| A2 Music / SFX<br>Clips snap to block boundaries by default (Snap on); Alt-drag disables snapping |

- **MVP editing operations only:** trim (drag edges), split (S at playhead), reorder blocks, replace asset, adjust clip volume, delete. No keyframing, no multi-cam, no nested sequences.
- Block boundaries are visible as labelled segments; dragging a clip across a boundary asks whether to move it or extend it - it never silently reassigns narration.
- **Motion graphics (Stage 8)** live on V2 as preset clips: lower-third, stat / chart card, tier list, quote card, list build, caption-style callout. Each preset exposes only text, accent colour, and in/out animation (3 options). No freeform canvas in MVP.
- Graphics presets use the project's style kit (Section 7.1) so a video looks consistent without design work per clip.
- Autosave plus named checkpoints: "Save checkpoint" in the Timeline menu; checkpoints list under Edit > History.
- Performance rule: timeline must stay responsive with 200 clips; thumbnails are lazy-loaded filmstrips, waveforms are pre-rendered images, not live-drawn.

# 7. Captions, sound, style kit, and Preview

## 7.1 Project style kit (set once, used everywhere)

Font (from a licensed bundled set of 8), caption colour pair, accent colour, logo (optional upload), graphics preset theme. Set at first visit to Captions; editable later, changes re-style captions and graphics without re-rendering until the user renders.

## 7.2 Captions

- Generated from the corrected transcript. Styles: **Standard** (2 lines, bottom), **Word-by-word highlight**, **Bold centre** (Shorts-style). Live preview on the monitor for each.
- Controls: font size (S/M/L), position (bottom / centre), background (none / bar / box), max characters per line (default 42).
- Caption text is edited in the transcript editor (Section 5.3), not a second editor here. This screen controls look only - the UI says so, with a link back.
- Burn-in vs sidecar: MVP burns captions into the main render and also exports .srt and .vtt. The Shorts renders use Word-by-word by default.

## 7.3 Sound

- Music: licensed library list with mood tags, or upload. One music track in MVP. Volume slider plus **Auto-duck under voice** toggle (default on, -12 dB).
- SFX: whoosh / pop / ding presets attachable to graphics clips from the Timeline inspector. No SFX browser in MVP.
- Final mix check: loudness meter on Preview showing integrated LUFS; warn if outside -16 to -13 LUFS.

## 7.4 Preview screen

- Full-quality proxy playback (720p proxy rendered by Remotion in-browser preview where possible, falling back to server proxy). Playhead syncs with timeline selection.
- Honesty rule: a label states "Preview is a proxy. Final render may differ slightly in graphics sharpness." if any element is proxied.
- Comment-free in MVP: no timestamped comments. Review happens via checkpoints and proposal history.

# 8. Render and Publish pack

## 8.1 Render screen

| Setting | MVP values |
|---|---|
| Main video | 1080p, 16:9, MP4 (H.264), 30 fps. 4K shown as "Later" - disabled, not hidden, with no price promised. |
| Also render | Checkboxes: 2 Shorts / Reels cuts (9:16, auto-selected blocks, user can change selection), 1 square (1:1) cut - off by default |
| Cost | Exact credit cost computed from duration and outputs, shown before Queue render. No render starts without a click on a button carrying that cost. |
| Queue | Position, elapsed, estimated remaining from recent jobs of similar length (labelled as estimate), Cancel before 10% progress with full credit refund; after that, no refund - stated before queueing. |

- Render failures show the failed stage (captions / graphics / encode), a Retry button that reuses completed stages, and credits automatically refunded for the failed portion. Support reference code is shown and copyable.
- Completed renders list: thumbnail, duration, size, Download, Re-render with same settings.

## 8.2 Publish pack

- Generated with the render, one ZIP plus individual downloads: main MP4, Shorts MP4s, 3 thumbnail options (frame + title text using style kit, user picks / edits text), title (3 options, pick or edit), description (with chapters from block titles and asset credits), hashtags, .srt / .vtt.
- Thumbnail editor is minimal: pick frame, edit 2 lines of text, pick accent. No layers, no stickers in MVP.
- No direct posting in MVP. Buttons say "Download pack" - the UI never implies it posted somewhere it did not.

# 9. States: empty, loading, error, and credits

## 9.1 Rules for every async stage

| State | Required behaviour |
|---|---|
| Empty | Say what belongs here, why it matters in one line, and give exactly one primary action. Never a blank panel. |
| Loading under 2s | Skeleton of the real layout, no spinner-only screen. |
| Loading over 2s | Named progress ("Transcribing block 3 of 8") via Job centre + inline indicator. Cancellable where backend allows. |
| Partial failure | Successful blocks stay. Failed blocks listed individually with Retry - never fail the whole stage for one block. |
| Full error | Plain-language cause, what was / was not charged, one Retry action, support code. Raw stack traces never reach the UI. |
| Offline / reconnect | Banner "Connection lost - edits are saved on this device and will sync". Editing continues locally; generation and render buttons disable with reason. |
| Session expired | Modal on next action, sign in again in place, zero edit loss. |

## 9.2 Credits and paywall states

- **Low credits:** pill turns amber under 20% of typical project cost; warning appears on the cost button, not as a pop-up.
- **Insufficient credits:** the action does not start. Sheet shows cost, balance, shortfall, and Buy / Upgrade actions. Partial generation (some blocks only) is offered where it makes sense.
- **Free / trial limits** (if a trial exists - see Pricing doc): watermark and resolution limits, if any, are stated on the Render screen before queueing, never discovered on the output file.

## 9.3 Confirmations that must exist

Delete project, delete take, replace approved script narration, enable YouTube opt-in, cancel a render after 10%, spend over a user-set credit threshold (setting, default off).

# 10. Accessibility and responsive behaviour

## 10.1 Accessibility - target WCAG 2.2 AA

- Full keyboard operation: stage rail, script blocks, proposal cards, and timeline trim / split / reorder all reachable and operable without a mouse. Visible focus ring (2px, accent colour, 3:1 contrast) on every interactive element.
- Contrast: body text 4.5:1 minimum, large text and UI controls 3:1. Stale / error / success are never signalled by colour alone - icon plus text label.
- Screen readers: stages are a labelled navigation landmark; job progress uses aria-live polite announcements ("Render 60 percent"); proposal cards are dialogs with labelled Apply / Dismiss actions; video preview has play / pause and time announced.
- Captions for all instructional and sample audio in the product itself (voice samples, help videos).
- Motion: respect prefers-reduced-motion - disable graphics preset previews' animation and all non-essential transitions.
- Targets: 40x40px minimum touch target; timeline clip handles have an enlarged hit area beyond their visual edge.
- Forms: every field labelled, errors tied to fields with text (not colour), and announced. Time estimates and credit costs are real text, not tooltips only.

## 10.2 Responsive behaviour

| Width | Behaviour |
|---|---|
| 1280px and up | Full layout: stage rail + canvas + inspector. |
| 900-1279px (tablet / small laptop) | Inspector becomes a slide-over panel. Timeline remains fully editable. |
| 600-899px | Stage rail collapses to a top stage stepper. Timeline editable for trim / reorder only; split and graphics editing prompt "best on a larger screen" but do not block trim. |
| Under 600px (phone) | Review mode: dashboard, script read / approve, proposal Apply / Dismiss, preview playback, render status, publish download. Timeline and transcript audio-editing show a clear "Open on desktop or tablet to edit" state - content is visible, not hidden. |

No horizontal scrolling at any breakpoint except inside the timeline strip itself. Preview monitor always keeps 16:9 (or project ratio) with letterboxing.

# 11. Open decisions and build checklist

Each item below is one decision. Build proceeds on the recommended default if no decision is recorded before the spike ends.

| Decision | Options and recommended default |
|---|---|
| OD-1: 9:16-only projects at creation | A) 16:9 only in MVP, Shorts derived (recommended). B) Allow native 9:16 projects in MVP. |
| OD-2: Timeline chat edit box in MVP | A) Ship in MVP, proposal-only (recommended if spike validates reliability). B) Defer to v1; MVP uses buttons / menus only. |
| OD-3: Recording length per take | A) Per-block recording only (recommended). B) Add one full-length take mode in MVP. |
| OD-4: Thumbnail editor depth | A) Frame + 2 text lines (recommended). B) Add background removal for a face cut-out in MVP. |
| OD-5: Phone editing scope | A) Review / approve only under 600px (recommended). B) Full timeline editing on phone in MVP. |

### Build checklist (UI)

- Shell, stage rail with 4 stage states, autosave indicator, job centre, credits pill.
- Dashboard with Continue, search, card actions, new-user empty state.
- Script blocks with proposal diffs and approve / stale logic.
- Voice: record, upload, AI voice, cleanup A/B, transcript editor with two labelled modes.
- Visual plan rows with licence labels and the YouTube opt-in gate.
- Timeline with the 6 MVP operations, graphics presets, checkpoints, 20-action history.
- Captions styles, style kit, sound ducking, LUFS warning, proxy-labelled preview.
- Render queue with pre-shown cost and refund rules; publish pack with ZIP + individual files.
- All Section 9 states implemented per stage; WCAG 2.2 AA audit before MVP freeze; responsive breakpoints tested at 1280 / 1024 / 768 / 390px.

> **Definition of done for this spec:** a designer can produce every screen above without inventing a state, and a developer can name the component, its props, and its empty / loading / error behaviour from this document plus the API Spec.